

Fifty years of research have deepened uncertainties in global irrigation water withdrawals

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For five decades, refining models and datasets has been regarded as central to reducing uncertainty in global irrigation water withdrawal estimates. However, whether these efforts have succeeded remains unclear. Here we compile 1,624 global estimates from 65 studies published between 1974 and 2025 and evaluate them across 360 analytical pathways, spanning a comprehensive range of plausible statistical designs. Our results indicate that uncertainty in global irrigation water withdrawals has not declined: only 17% of pathways show any reduction over time, falling to 10% once projections to 2100 are excluded. The most common outcome is an increase in uncertainty (40% of pathways), followed by an unstable trend (34%). Progress may therefore require the field to renounce the goal of reducing uncertainty and instead double down on what it has historically done best: expanding it, until uncertainty reaches the boundary of what current knowledge of global irrigation cannot exclude.

The quantification of the volume of water globally withdrawn by irrigated agriculture has been a key hydrological goal since the Second World War¹. Initially derived from statistical exercises and mostly produced through global hydrological models after 2000, global irrigation water withdrawal (IWW) estimates shape our understanding of irrigation's impact on food, water and environmental security. Key assumptions—that irrigation accounts for 70% of freshwater withdrawals and 40% of food production², exceeds the blue water planetary boundary^{3,4} and depletes groundwater beyond sustainable limits⁵—are all grounded in IWW data. IWW figures also have strong policy implications: the Food and Agriculture Organization of the United Nations (FAO) use them under Sustainable Development Goal 6.4 ('increase water-use efficiency across all sectors and ensure sustainable withdrawals')⁶, whereas the Intergovernmental Panel on Climate Change (IPCC) integrate them into projections of land-use change and socio-climatic scenarios^{7,8}, shaping narratives about future water availability.

IWW estimates were deemed highly uncertain since the beginning. L'vovich¹ and Shiklomanov⁹ identified the lack of empirical data

due to uneven regional coverage as a major obstacle to accurate water accounting. Gleick¹⁰ noted that these gaps were often filled with computed data, introducing pitfalls such as hidden assumptions, inadequate attention to uncertainties, variability in hydrological data and illusory precision. The FAO affirmed that modelled values, although valuable, 'cannot replace local measurements'¹¹. The issues of IWW numbers at the beginning of the 2000s were so obvious that caution was advised when using them because, jointly with industrial and domestic withdrawals, they were 'the least reliable and most inconsistent of all water-resources information'¹².

Reducing uncertainties to narrow down the spread of IWW estimates and bring them closer to the 'true' value was therefore seen as essential to advancing the field. This vision aligned with the notion of uncertainty as the antithesis of knowledge—something that research must seek to progressively eliminate¹³. Calls were issued to reduce uncertainties by producing more detailed maps of irrigated areas and crop types^{14–16}, improving historical weather reconstruction and reanalysis products¹⁷, intensifying scientific collaborations⁹ and adopting

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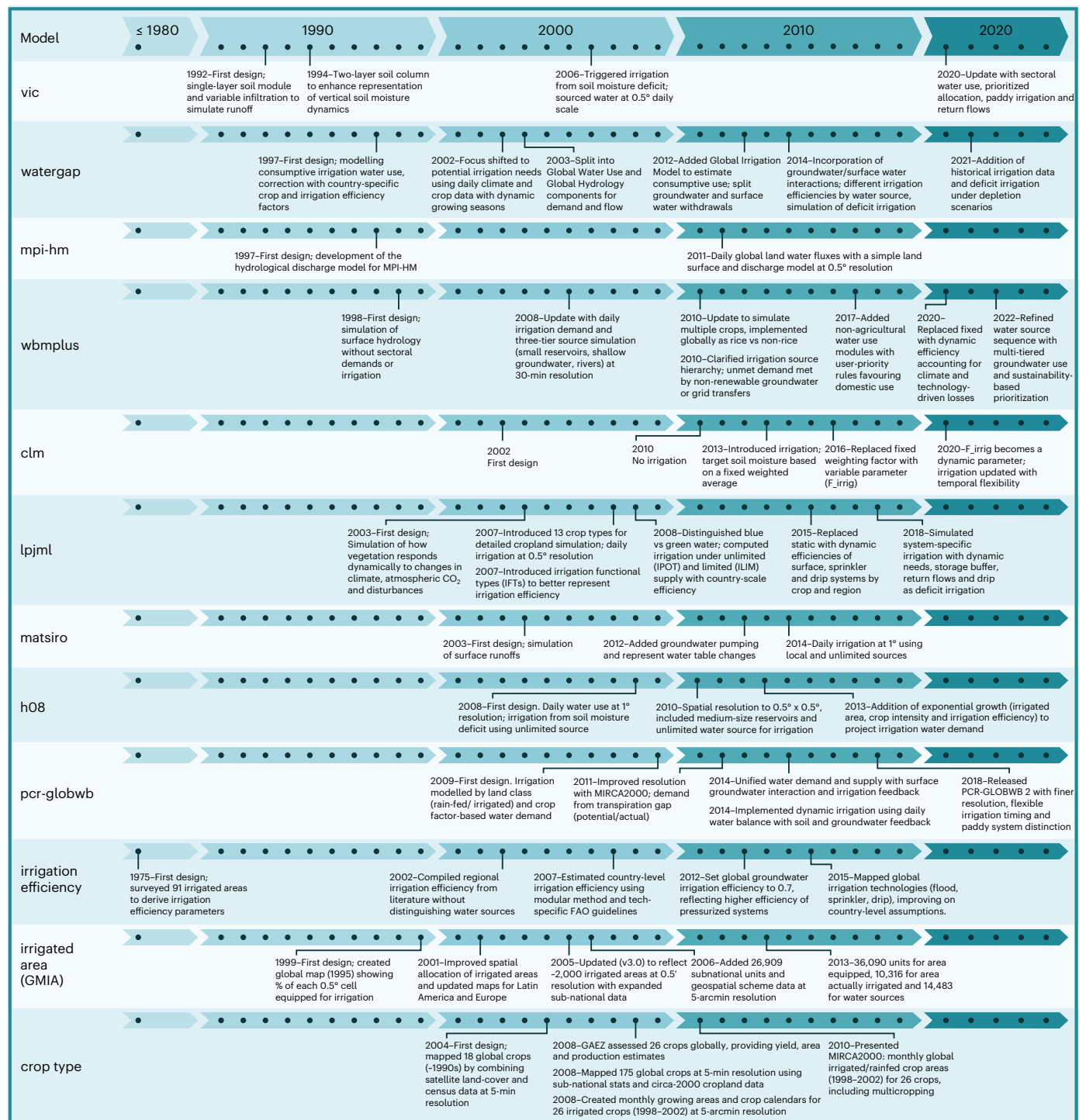


Fig. 1 | Summary of model and irrigation dataset updates over the last 50 years. The full dataset with the definition of the model names can be found in Puy et al.²⁸.

high-resolution modelling, remote sensing or machine learning^{18–20} (Supplementary Text 1). The hydrological community responded accordingly: country-level and basin-scale estimates based on census statistics and fixed irrigation coefficients were replaced by spatially explicit, gridded models with remotely sensed irrigated areas, cropping calendars, historical climate forcing and coupled land-surface or hydrological models (Fig. 1). To date, the consideration of uncertainties as a transient stage to be overcome through finer-grained detail and more data remains the prevailing view in hydrological modelling^{21–23}.

Here we show that five decades of data and model refinement have done little to reduce uncertainties in IWW estimates. Using a comprehensive statistical approach to explore how results vary across

analytical pathways (specific, end-to-end sequences of methodological choices made to collect, process and analyse data)²⁴, we find that the strongest signal is not a reduction, but an increase in uncertainty. Paths showing a consistent decrease in uncertainty are very few and circumscribed to specific analytical settings. Our results raise broader questions about the future of global irrigation research and whether progress should remain tied to the pursuit of numerical precision, even when it has proven historically elusive. Building on a large tradition of hydrological studies on uncertainty^{25–27}, our study shifts the focus from chasing a narrower spread to deliberately enlarging uncertainty until it reflects the true breadth of plausible estimates, to advance understanding and support more flexible, resilient water science and policy.

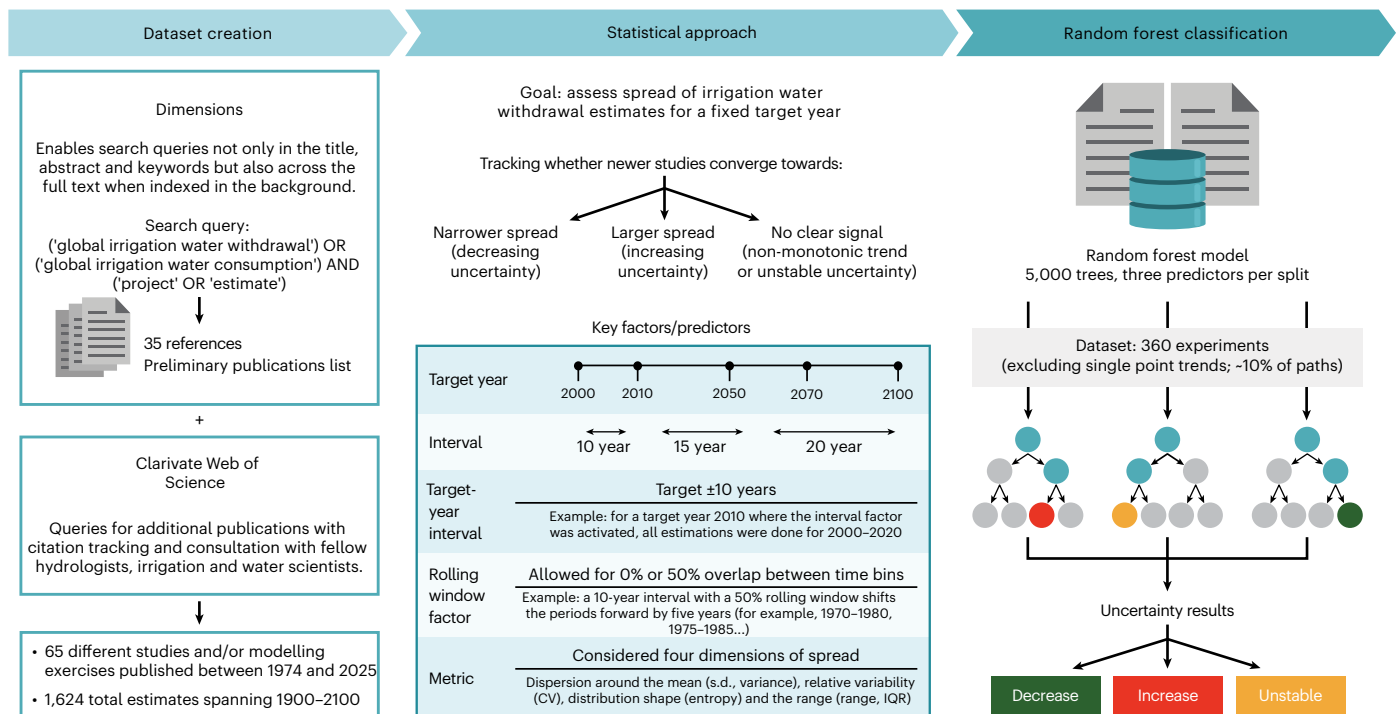


Fig. 2 | Summary of our research design. In the ‘Statistical approach’ tab, s.d., CV and IQR stand for standard deviation, coefficient of variation and interquartile range, respectively. Methods provides further information. Credit: Karen Thiebes under a Creative Commons licence CC BY.

We used the Dimensions and Web of Science databases to compile 1,624 global IWW estimates (1900–2100) from 65 studies published between 1974 and 2025 (Fig. 2 and Methods), most of which did not conduct an uncertainty (UA) or an uncertainty and sensitivity analysis (UA/SA) (Fig. 3a,b). Many studies called to reduce uncertainties through finer-grained irrigation maps, higher-resolution data or more realistic models, but few claimed improvement over earlier work, even though one-third had the assessment of uncertainty as a key research goal (Supplementary Text 2)²⁸. A large majority of estimates were reported in journal articles, with a smaller portion presented in documents by the FAO, the Organisation for Economic Co-operation and Development (OECD) or the World Resources Institute. Approximately 85% of the estimates were made for the year 2000 and later, with 2010 (97), 2020 (121), 2050 (152), 2070 (148) and 2100 (121) being the years with more estimates (Fig. 3c). Overall, about 25 models were used in the production of estimates, with only eight studies not providing information about the model used (Fig. 3d).

Given the many degrees of freedom in assessing the effects of fifty years of research on the uncertainty of IWW estimates, we explore multiple analytical pathways within a comprehensive statistical approach to evaluate whether uncertainty has increased or decreased over time (Fig. 2). We account for the most plausible alternatives in the design of the analysis, especially with regards to the factors that could influence the outcome: the selection of the target year of the IWW estimation, the criteria for binning studies over time and the metric adopted to calculate the spread of IWW estimates, which we use as a proxy for uncertainty. We combine all levels for these factors in a full factorial design, produce 360 possible statistical evaluations and explore them all to assess how findings vary across them (Fig. 3e, Supplementary Fig. 1 and Methods). These array of choices can be pictured as a ‘garden of forking paths’²⁹, where each crossroad represents a diverging direction that the analyst needs to navigate in the way to produce the scientific inference (Box 1).

Our approach minimizes the risk of generating volatile inferences (inferences that seem robust under a particular path but may shift when an alternative, equally plausible path is considered) by exploring all forking paths of the garden at once. Our exercise thus connects with a tradition concerned about the many reasonable but contingent

analytical choices available to researchers during the research design stage, an approach still incipient in hydrological and climate modelling but influential in psychology²⁴, political science³⁰, medicine³¹ and neuroscience³² that goes by the name of ‘modelling of the modelling process’³³, ‘multiverse analysis’²⁴ or ‘vibration of effects’³⁴.

We assume that a steadily decreasing spread of estimates over time indicates reduced uncertainty. In contrast, a non-monotonic or increasing spread suggests that research has either had an inconsistent impact on uncertainties or has actually increased them (Figs. 2 and 3f). Because a smaller spread may exclude the ‘true’ value while a larger spread may end up encompassing it (and ‘validation’ of global IWW estimates is beyond reach³⁵), we refrain from interpreting these trends in terms of increasing or decreasing accuracy.

Little evidence for research having reduced uncertainties

The assumption that progressively refining models and datasets has reduced the uncertainty in IWW estimates is the least supported by available data. Only 17% of analytical paths suggest a decline in uncertainty over time (Fig. 4a,b). The notion that 50 years of research have actually increased uncertainty has the strongest support (40%). Because one-third of the analytical paths lead to scenarios where uncertainty shows an unstable trend (34%), the most evidence is for research having had no consistent effect in stabilizing or reducing uncertainty. These results are not conditioned by structural differences in data availability such as the smaller number of models and estimates for earlier decades: a directional analysis testing whether the direction of change in uncertainty aligns with changes in the number of models, studies and estimates over time only finds weak and inconsistent associations (Supplementary Fig. 2), thus suggesting that uncertainty has not evolved as a simple function of research volume.

Research reduced uncertainty for 2100 but amplified it for other years

We use a random forest model consisting of an ensemble of 5,000 decision trees to assess the relative importance of each methodological

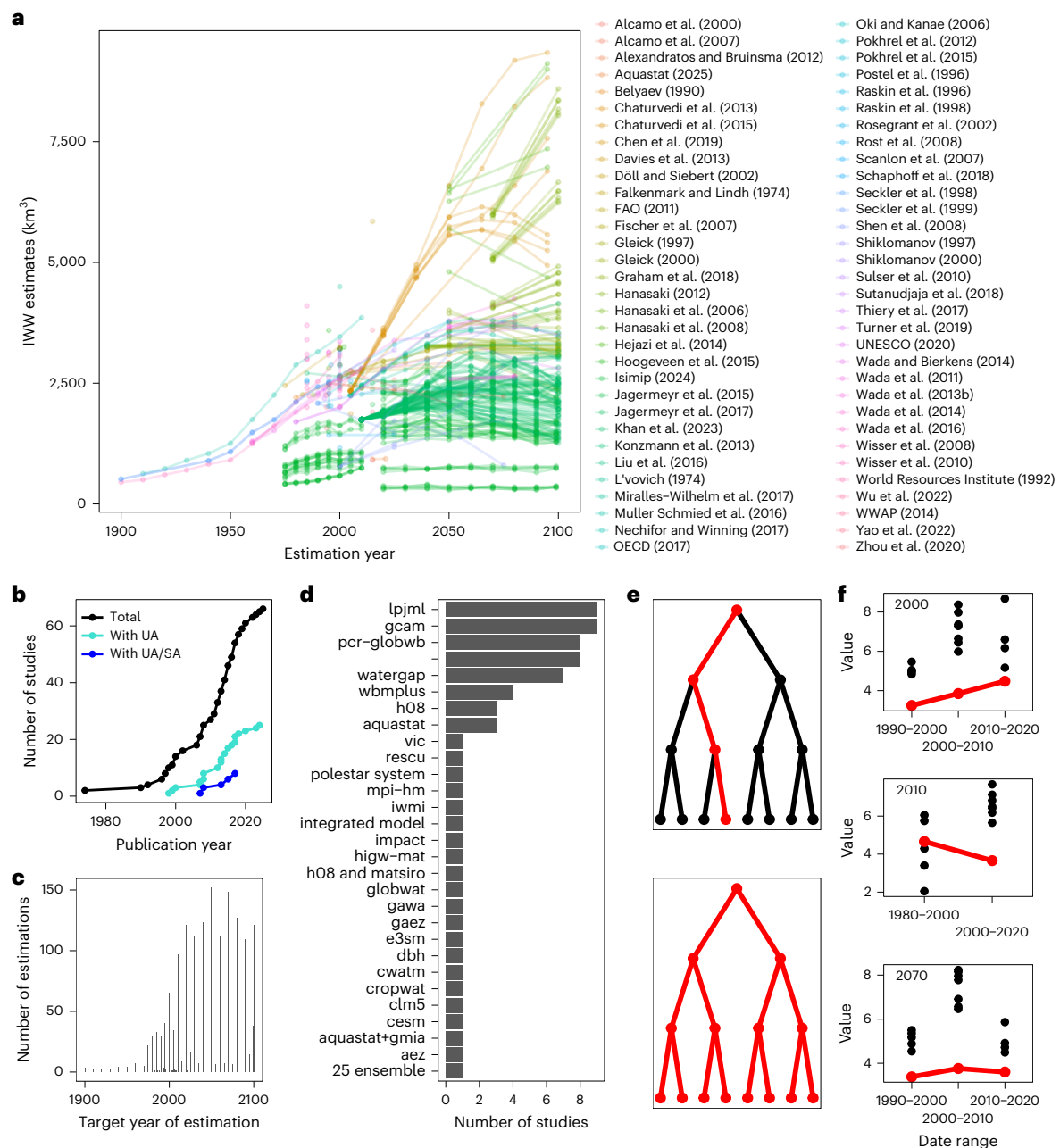


Fig. 3 | Estimations of IWW. a, Line plot of all IWW estimates included in the analysis. Puy et al.²⁸ provide the full references. UNESCO, United Nations Educational, Scientific and Cultural Organization; WWAP, United Nations' World Water Assessment Programme. **b**, Cumulative count of studies estimating IWW over time. UA and SA stand for uncertainty analysis and sensitivity analysis, respectively. **c**, Number of estimates for each target year. **d**, Bar plot showing the number of studies per model. **e**, Illustration of the multiverse analysis. The top graph depicts a traditional research process occurring over a single analytical path (in red), whereas the bottom plot illustrates a research design

where all plausible paths and methodological crossroads are explored at once. **f**, Three hypothetical outcomes of our statistical design. The dots represent IWW estimates for a given target year (the number on the top left side of each plot) from studies published in specific time intervals (the x axis; 1990–2000, 2000–2010 and 2010–2020 in the top plot). Red dots indicate the measure of spread (for example, standard deviation) for each period, whereas the red line represents the trend. An increasing (decreasing) trend suggests growing (declining) spread in IWW estimates for a given target year over time. The bottom plot illustrates an example of an unstable or non-monotonic trend.

decision in shaping the perceived uncertainty trend (Fig. 2 and Methods). The selection of the target year is the most influential methodological choice (Fig. 4c). Most analytical pathways suggest a progressive reduction in the uncertainty of IWW estimates for the year 2100, whereas for other target years increasing or unstable trends dominate (Fig. 4d).

The decline for 2100 probably stems from the removal of extreme socio-climatic scenarios in recent hydrological studies—scenarios that widened estimate ranges (and thus led to increased uncertainty)

but were deemed implausible as research advanced. For instance, the highest IWW estimates for 2100 (8,000–9,000 km³; Fig. 3a) come from a 2012 study combining Shared Socioeconomic Pathway 3 (SSP3; low climate mitigation) with Representative Concentration Pathway 8.5 (RCP8.5; 3.5–5.4 °C warming) (Table 10 in ref. 36) and a 2014 study assuming a population of 14 billion by 2100 (ref. 37). However, SSP3–RCP8.5 is not a plausible scenario in the IPCC sixth assessment (Fig. 6 in ref. 38) and is absent from recent irrigation studies⁷, whereas the latest United Nations³⁹ projection of population for 2100 is 10.4 billion (95%

BOX 1

The garden of forking paths

Brought into scientific discourse by Gelman and Loken²⁹ after borrowing the term from Borges⁷⁶, the ‘garden of forking paths’ metaphor highlights the methodological flexibility existing in the framing and implementation of a given research question, with each path reflecting a plausible analytical trajectory. Here we conduct a full exploration of 360 different analytical paths formed by combining different target years, ways to bin studies and uncertainty metrics. The examples below illustrate our approach:

First path: has uncertainty (for example, the standard deviation) in estimating irrigation water withdrawals for 2050 decreased, given the trends observed across studies published between 1990–2000, 2000–2010 and 2010–2020?

Second path: has uncertainty (for example, the range) in estimating irrigation water withdrawals for 2000–2010 decreased, given the trends observed across studies published between 2000–2010, 2005–2015, 2010–2020 and 2015–2025?

Third path: (...)

:

360th path: has uncertainty (for example, the entropy) in estimating irrigation water withdrawals for 2050 decreased, given the trends observed across studies published between 1980–1990, 1990–2000, 2000–2010 and 2010–2020?

CI: 8.9–12.4). With these extreme scenarios no longer included in recent works, estimate ranges for 2100 have narrowed relative to older ones, driving the observed predominant decrease in uncertainty. Notably, when 2100 is excluded as a target year in our statistical analysis, the proportion of pathways supporting a decline in uncertainty drops to 10%, with approximately 45% indicating an increase and 40% suggesting an unstable trend.

For target years between 2000 and 2070, paths leading to an increase in uncertainty outnumber those showing a decrease, with the difference reaching a factor of ten for historical years (for example, 2000, 2010) (Fig. 4d). This means that having access to historical data for calibration and evaluation—such as crop intensity data (available since about 1995 (ref. 14)), data on irrigated areas (since 2000 (ref. 40)) or country-level IWW data (since 1994 (ref. 2))—has not clearly narrowed the spread of IWW estimates. An exploratory analysis of variance suggests that > 80% of this spread is driven by how irrigation is modelled (with a minor contribution of climate forcing and socio-climatic scenarios; Supplementary Text 3 and Supplementary Fig. 3), indicating that refining models has done little to mitigate the uncertainty arising from differing model assumptions and representations of irrigation. For instance, 67 out of 102 assumptions attested in nine global irrigation models are unique to a single model⁴¹, and changes in the formalization of irrigation efficiency and crop evapotranspiration translate into IWW values that differ by more than an order of magnitude^{42,43}.

The continuous update of datasets does not seem to have facilitated convergence of IWW estimates either. The Global Map of Irrigated Areas (GMIA) is a case in point: developed in 1999 to inform on the area equipped for irrigation around 1990 at a 0.5° × 0.5° resolution⁴⁴, it was updated in 2005 (v.3.0) to reflect irrigated areas around 2000 with a higher resolution (0.5° × 0.5°) and expanded statistics for 10,825 subnational units⁴⁰. In 2013 (v.5.0) it was further enriched with about 25,000 new subnational units, information on the source of irrigation water and the distinction between area equipped for irrigation and the area actually irrigated⁴⁵ (Fig. 1). Since its inception, the GMIA has been used to parametrize irrigated areas in nearly all global models of irrigation agriculture. These models are highly sensitive to the GMIA

data as irrigation extent is one of the most important parameters influencing IWW^{47,48}. Given this degree of reliance and influence, our results suggest that the GMIA updates have not helped models reduce the uncertainty in IWW estimates—they may even have reinforced model discrepancies, leading to larger uncertainties with each update. Examples where finer-grained detail in spatially distributed datasets leads to larger output uncertainties abound in hydrology and are explained by higher-resolution expanding errors if other modelled processes are poorly constrained or known^{47,48}.

The definition of spread conditions the perceived trend

The calls to reduce uncertainty through better models and datasets have assumed that the spread of IWW estimates is an objective feature of the data (Supplementary Text 1). Our results indicate that it is a property shaped by the act of measurement itself: changing the spread metric can shift the perceived uncertainty trend (Fig. 4c). This variability arises because uncertainty is multi-dimensional and each metric captures only a specific aspect of it: variance and the standard deviation (s.d.) focus on the dispersion around the mean, the coefficient of variation (CV) and the range account for relative dispersion (to the mean and the extremes), entropy focuses on the evenness of the distribution and the interquartile range (IQR) accounts for the middle 50% of the data.

Figure 5a–c illustrates this phenomenon. In Fig. 5a, the CV suggests decreasing uncertainty as relative dispersion declines over time, whereas the s.d. indicates an unstable trend because the spread of IWW estimates peaks in studies published between 2010 and 2019 before decreasing again in 2020–2025. Entropy suggests increasing uncertainty because IWW estimates become more evenly distributed over time and hence it is less and less clear which value dominates. In Fig. 5b, the s.d. decreases from 2010–2019 to 2020–2025 despite the larger spread in the later period because the presence of a dense cluster of ~ 60 estimates near the central value (~ 1,740 km³ (ref. 7)) reduces the overall dispersion. A similar pattern between the IQR and the variance can be observed in Fig. 5c. Overall, 58% of our paths produced two possible trends and 7% three possible trends just by changing the spread metric.

The lack of clarity in global irrigation modelling on which dimension of uncertainty should be reduced highlights a fundamental gap in defining progress through diminishing uncertainty. Should we minimize variability, extreme values or enhance convergence around a central estimate? These goals are not necessarily compatible and there may be trade-offs. For instance, smoothing extreme values linked to specific IWW models may be possible by adopting a single global IWW model (akin to an ‘irrigation’ version of the Community Hydrological Model⁴⁹), but the middle 50% of the data (as measured by the IQR) may remain unchanged. Adopting higher-resolution rainfall data may reduce the spread of IWW estimates as measured by the s.d., but amplify their range by better capturing extreme values and precipitation variability⁵⁰. Agreeing on a definition of irrigated area (for example, by addressing reporting inconsistencies²²) may narrow the range of IWW estimates but could also make it harder to identify the most probable value if estimates become more evenly distributed. Thus, how uncertainty is framed is crucial for its effective treatment⁵¹. Overall, research has mostly reduced uncertainty only if the spread is defined as the middle 50% of the data. If uncertainty is measured by the full range, entropy or the coefficient of variation, then research has clearly expanded it (Fig. 4d). This means that uncertainty is a by-product of scientific enquiry and hence not reducible to a transient epistemic gap.

Research has increased uncertainty in the long run

Scientific trends may be distinguishable over short time spans if driven by the adoption of new methods or over decades if shaped by fundamental shifts⁵². The impact of research on IWW uncertainty is much clearer over intervals of 15–20 years, with trends indicating an increase

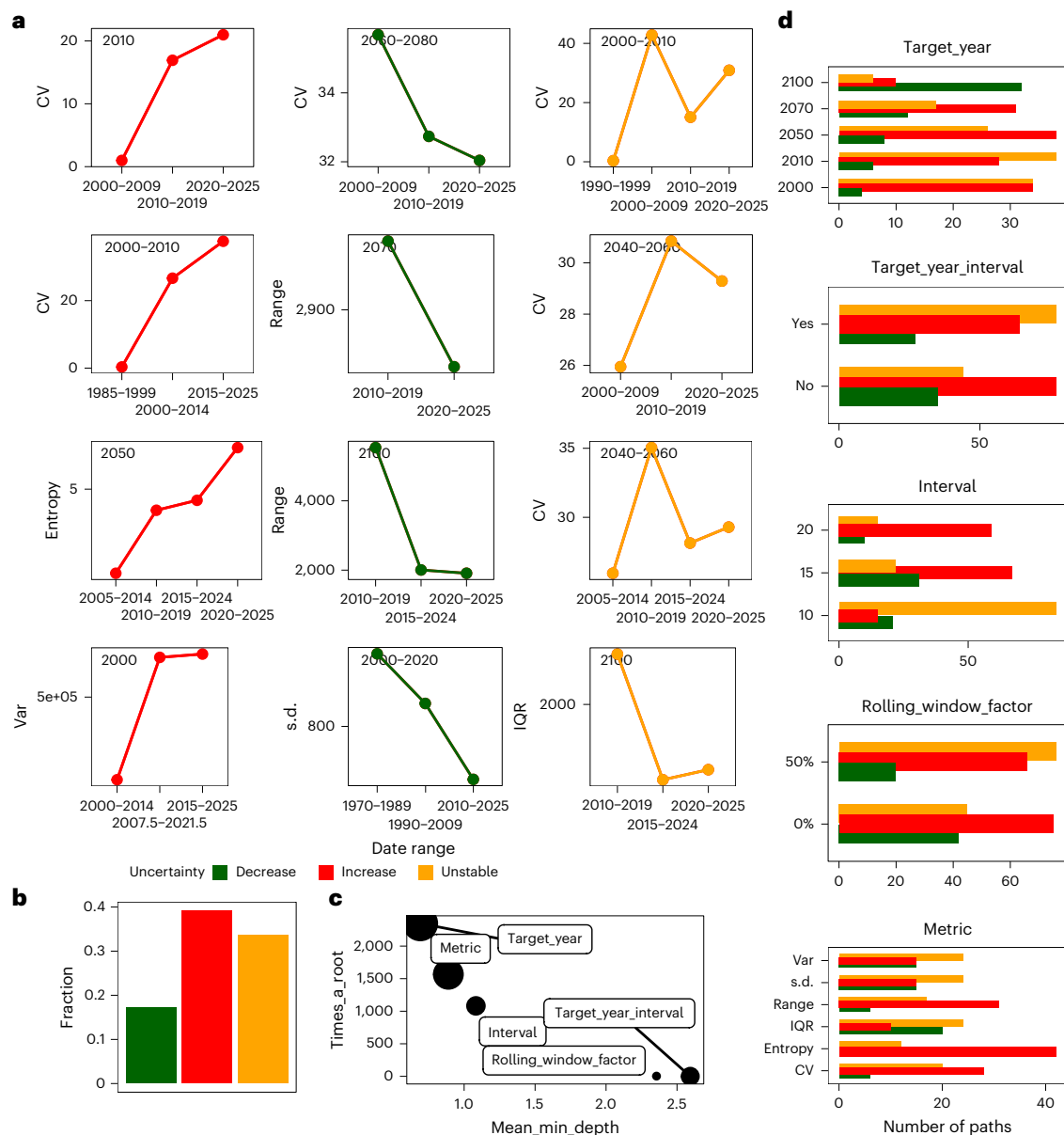


Fig. 4 | Results of our statistical approach. a, Examples of analytical paths where uncertainty increases, decreases or is unstable over time (columns from left to right). Labels at the top left corner in each plot represent the target years or target-year intervals for which IWW estimates were made. The x axis represents the publication years of the studies yielding the IWW estimates, whereas the y axis shows the metric used to appraise the uncertainty. Var, variance; s.d., standard deviation; CV, coefficient of variation; IQR, interquartile range.

b, Fraction of paths suggesting that uncertainty in the estimation of IWW has decreased, increased or been unstable over time. **c**, Importance plot. Factors with higher times_a_root and lower mean_min_depth values have a higher influence because they are used more often and earlier in the split of decision trees. The larger the node, the more the factor is involved in decision nodes. **d**, Bar plot showing the output of the paths as a function of each uncertain methodological factor.

in uncertainty outnumbering those indicating a decrease by a factor of two to six (Fig. 4d). With shorter intervals (for example, 10 years), our statistical approach produces more unstable trends because the number of year bins increases and hence the trend captures more short-term variations and noise. Notably, analytical paths indicating an increase in uncertainty span a broader range of intervals and target years than those indicating a decrease or unstable trends, highlighting the temporal strength of the signal (Fig. 5d).

Even if the adoption of machine learning, remote sensing or similar technological innovations shift the predominant trends observed in the last five decades^{21,22}, the 15–20-year lag in detecting consistent patterns suggests that unquestionable reductions in IWW uncertainty are unlikely to be seen anytime soon. This reveals a mismatch between the slow pace of epistemic progress (if framed as a quest for numerical

precision) and the urgency of irrigation-related challenges. Food and water security, climate adaptation and freshwater sustainability are pressing issues that partially rely on global models for guidance^{53,54}. Linking progress to the pursuit of higher numerical precision despite historical trends suggesting it is unlikely (and its effects belated) may undermine the social relevance of models and invite decision-making to outpace them.

Discussion

Fifty years of increasingly detailed datasets and models aimed at reducing uncertainty in the estimation of global IWW have largely had the opposite effect. The volume of water globally used for irrigation may thus be an example of quantification that becomes less accurate as understanding deepens. Similar patterns can be found elsewhere:

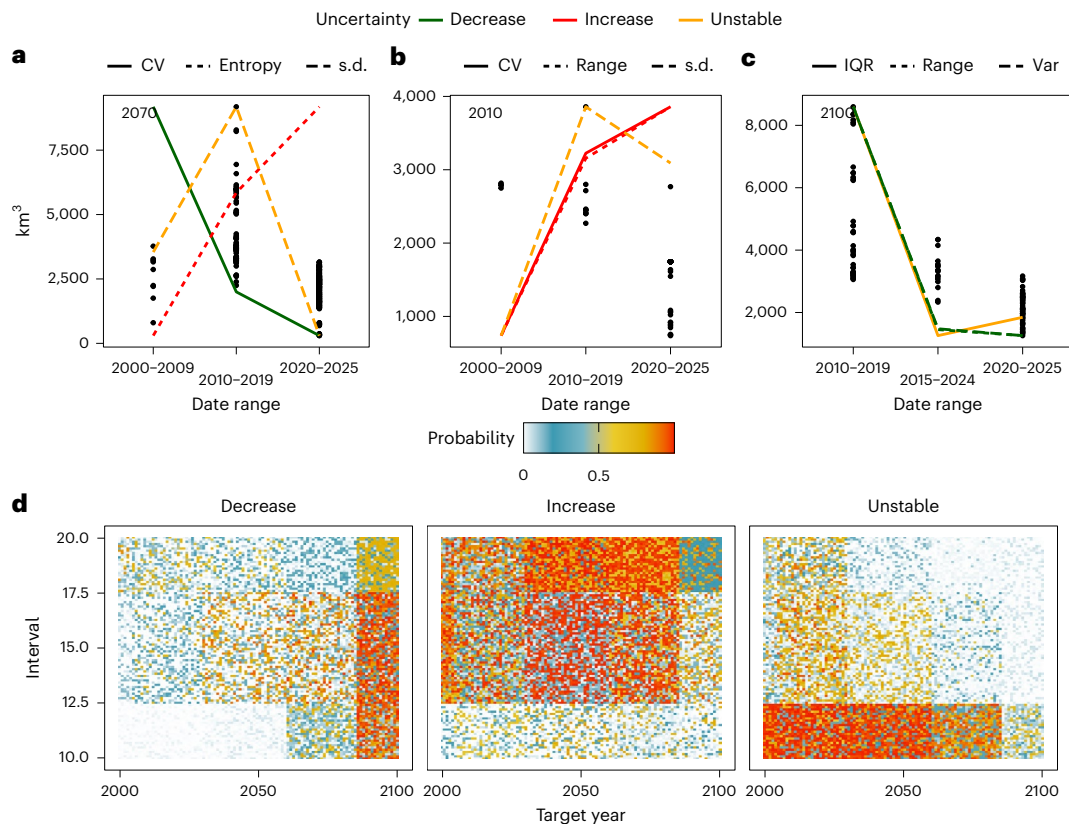


Fig. 5 | Uncertainty in the metric and results of the random forest model.

a–c, Examples of analytical paths where a change of metric (with all other factors kept fixed) modifies the uncertainty trend. Dots indicate IWW estimates from studies published in years within each time bin (x axis) for 2070 (**a**), 2100 (**b**) and 2100 (**c**). The lines show the evolution of uncertainty over time as a function of the metric used, with green, red and orange indicating decreasing, increasing

and unstable trends, respectively. The trends of the metrics have been rescaled to overlap with the underlying data. **d,** Grid plot showing the interaction between target year and the interval in the random forest model. Warmer (colder) colours indicate higher (lower) probability of the interaction producing a 'decrease', 'increase' and 'unstable' uncertainty trend. Each pixel reflects the averaged probability from all the trees in the forest.

for example, the confidence level associated with the uncertainty in climate sensitivity (1.5–4.5 °C; the average global warming following a doubling of atmospheric CO_2) went from 95% interval in the first IPCC report (1990) to 66% interval in the fifth IPCC report (2014), despite the use of increasingly sophisticated multi-model ensembles⁵⁵. The claim that irrigation withdraws 70% of freshwater and produces 40% of global crops, uttered over the past 50 years by more than 3,000 studies, has recently turned into much broader ranges (45–90% and 18–50%, respectively)². Evidence also suggests that global IWW estimates will become even more uncertain if parametric and structural uncertainties are fully accounted for in the simulations⁴².

When the phenomenon under study is complex, research tends to multiply ambiguities by bringing to light its multifaceted richness, a process that sometimes exposes the indeterminacy of the entity itself. In irrigation studies, progress has rendered the term 'irrigated agriculture' an open-ended reference (it can both include or exclude water harvesting/runoff systems and cultivated wetlands³⁵) and shown that the rainfed-irrigated divide discretizes what is a continuum of agricultural water-management practices of different intensity^{56,57}.

Larger uncertainties are valuable when they result from broadening the evidence base to better mirror real-world ambiguities. Namely, when global irrigation models account for the challenges in defining irrigated areas⁴⁶ (An 'area equipped for irrigation'? The area 'actually irrigated'? How can we meaningfully distinguish between rainfed and irrigation agriculture?), the existence of about 40 evapotranspiration equations and hence the lack of a 'true' one⁵⁸ or that many regions rely on traditional irrigation systems whose logics do not prioritize crop and water productivity⁵⁹. The value of uncertainties increases further if

the variance of IWW estimates is traced back through global sensitivity analysis to pinpoint which ambiguous assumption contributes most to overall uncertainty⁶⁰. If the dominant source can be reduced, uncertainty guides future research. If not, it clarifies the limits of knowledge and prompts reflection. Either outcome advances understanding.

A progressive reduction in uncertainty does not necessarily constitute epistemic progress because convergence may arise from growing model similarity and not from getting closer to the 'true' IWW value. This may happen when models increasingly share common components (without this overlap mirroring a better grasp on reality) or when an influential modelling group sets the standard for the representation of irrigation. For example, the GMIA parametrizes the extension of irrigation in almost all models despite the existence of at least four alternative and highly different datasets⁴⁶. The irrigation modules of PCR-GLOBWB, WaterGAP, WBMPlus and VIC are all similar and aligned to FAO's crop-evapotranspiration approach, whereas those of CLM4.5, H08, LPJmL and MATSIRO apply a common soil-moisture model (Supplementary Text 4). The overlap also extends to assumptions, with 35% of the 102 assumptions documented in models being adopted by at least two modelling teams⁴¹.

Progressive convergence of IWW estimates may signal real improvement only if preceded by a global uncertainty and sensitivity analysis (UA/SA) because only then can we be sure that it reflects robust inference rather than overlapping assumptions or structural similarities. However, only 24 out of the 65 studies compiled here (37%) propagated some type of uncertainty in the calculation of IWW, and almost all of them only assessed socio-climatic uncertainties (for example, by using different Global Climate Models, Shared Socioeconomic

Pathways and/or Representative Concentration Pathways) (Supplementary Text 5). Only eight (12%) combined an uncertainty with a sensitivity analysis, typically by moving one or two parameters with the one-at-a-time approach, known to perfunctorily explore the uncertainty space⁶¹ (Fig. 3b). We did not document any study performing a global UA/SA on the parameters involved in the calculation of IWW (for example, evapotranspiration, crop coefficient, irrigated area, precipitation). These results are in line with other studies that documented a poor appraisal of uncertainties in fields that rely on mathematical modelling^{62,63}, including hydrological modelling⁶⁴.

Our statistical analysis illustrates how a simple research question ('has uncertainty in IWW estimates decreased over the last 50 years?') produces different results when the sequence of methodological choices is expanded within reasonable limits. The results suggest that the most plausible answer to this question is 'no' and also clarify how each analytical path shapes the final inference. These findings have three key implications for the future of global irrigation modelling:

- (1) The foundational question of the field ('how much water is globally withdrawn by irrigation?') may not have a numerical answer. Five decades of model and data refinement have deferred this realization by retreating into technical complexity. Whereas future methodological advances may improve understanding of irrigation water withdrawals, our results show that earlier innovations widely regarded as promising for improving accuracy (for example, high-resolution⁶⁵, remote sensing⁶⁶) did not reduce uncertainty. This suggests the need for caution in assuming that new innovations will do so. For instance, the success of machine learning in reducing uncertainty for temperature projections⁶⁷ does not translate into irrigation easily: whereas temperature is an observable governed by first principles and dense measurements, irrigation agriculture is a socio-environmental construct made modellable through specific modelling choices: more data is unlikely to clarify how to delineate irrigated areas, conceptualize irrigation efficiency⁴³ or formalize different irrigation practices³⁵. Confronting this challenge is urgent given that the field will move further from pinpoint accuracy if it widely adopts global UA/SA⁴².
- (2) A moratorium on model and data expansion may improve understanding and allow us to fully embrace uncertainties⁶⁸. The fact that the pursuit of finer-grained data and models to reduce uncertainties has mostly led to the opposite result is not a failure but a misalignment between intent and outcome. However, continuing this trajectory without accounting for pre-existing uncertainties is unlikely to be productive: it risks compounding and propagating them through layers of complexity, further obscuring their nature⁶⁹. Global irrigation modelling will align better with empirical trends if it deliberately expands uncertainties; that is, if it aims at producing irrigation water withdrawal estimates so spread out that approach the bounds of what is plausible given current knowledge. Technically, this requires redirecting computational resources from model and data expansion towards comprehensive global UA/SA to explore the full space of assumptions. Conceptually, it means using models as tools to navigate uncertainty rather than as instruments to quantify³⁵.
- (3) Models should turn into 'gardens of forking paths'. When uncertainty is no longer an issue to address, models do not need to function as measurement devices executing one single analytical pipeline. They can be designed as 'gardens of forking paths', where each model component (for example, irrigated areas, crop schedules, onset of irrigation, soil maps) is a decision point with several plausible configurations. Such framework permits the integration of new datasets, assumptions and components as research advances and accommodates uncertainties as they emerge. Each path through the model now corresponds to a coherent but partial account of irrigation, with global UA/SA used

to traverse the full space of options at once. UA/SA turns from a post-hoc diagnostic to the primary objective, in agreement with uncertainty being the most persistent outcome of 50 years of research on global irrigation.

Recasting models from instruments of accuracy into instruments that expand uncertainty also changes what policy can ask of them. Rather than deliver a number around which policies should converge, global irrigation models would reveal what withdrawals current knowledge cannot rule out and the policy space they leave open across hydrology, agronomy and governance. Their value for policy-making would lie in disciplining judgement and exposing where decisions on water stress, irrigation-efficiency targets and food-production risk remain political. Such decisions would then return to democratic deliberation rather than being pre-empted by fragile numbers masquerading as evidence⁷⁰.

Progress in our understanding of global irrigation water withdrawal does not require the pursuit of more numerical accuracy, but the production of larger and more insightful uncertainty.

Methods

Creation of the dataset

We used the Dimensions database to compile a preliminary list of publications reporting on global irrigation water withdrawal estimates, either historical or projected. Unlike Scopus and the Web of Science, Dimensions enables search queries not only in the title, abstract and keywords but also across the full text when indexed in the background.

Our search query was ('global irrigation water withdrawal') OR ('global irrigation water consumption') AND ('project' OR 'estimate'), which returned 35 references. We used 'project' as a stem to capture multiple related forms, such as 'project', 'projected' or 'projection'. We included 'consumption' to capture studies conflating withdrawals with consumption, a known issue in the literature¹². We completed this list with additional studies retrieved after similar search queries on the Web of Science, citation tracking and informal consultation with fellow hydrologists and water scientists (Fig. 2).

For each reference, we retrieved the document in PDF and recorded the title of the study, the publication date, the irrigation water withdrawal estimates, the years for which the estimates were computed and the climate setting (if any). When the estimates were not made for a specific target year but rather for a period (for example, 2000–2010), we kept the middle year of the period as the target year. If the data was in a bar or line plot, we extracted the estimates using WebPlotDigitizer⁷¹. For geo-spatial data (for example, irrigation water withdrawals calculated at a spatial resolution of $0.5^\circ \times 0.5^\circ$; .nc4 files from the Inter-Sectoral Impact Model Intercomparison Project), we generated global estimates using the algorithm in Puy et al.²⁸ and retrieved estimates every 5–10 years (for example, 2010, 2015, 2020). Overall, we compiled 65 different studies and/or modelling exercises published between 1974 and 2025, collecting a total of 1,624 estimates spanning 1900–2100.

The statistical approach

We examined whether research has progressively reduced uncertainty in global irrigation water withdrawal estimates across a wide range of statistical designs (Fig. 2). This approach is known in psychology, economics, medicine or neuroscience as 'modelling of the modelling process'³³, 'multiverse analysis'²⁴ or 'vibration of effects'³⁴. It strengthens the statistical inference by addressing the problem of the researcher's degrees of freedom—the existence of many equally legitimate methodological paths in the framing of an analysis, which risks results being driven more by the specific analytical path chosen than by the true characteristics of the studied process²⁹. This latitude enables cherry-picking and unconscious bias and has contributed to the reproducibility crisis in science⁷².

We assessed the spread of irrigation water withdrawal estimates for a fixed target year, tracking whether newer studies converged

toward a narrower spread (decreasing uncertainty), a larger spread (increasing uncertainty) or if there was no clear signal (non-monotonic trend or unstable uncertainty). To ensure robustness in the results, we identified five key factors conditioning the design of our analysis and defined their most plausible levels based on the following rationale (Supplementary Fig. 1):

- (1) Target year. Past irrigation water withdrawal estimates (for example, < 2020) could be more constrained by data while predictions (for example, > 2025) could show greater variability as they are more dependent on untested climatic and socio-economic scenarios. This suggested that the selection of the target year could shape perceived uncertainty trends over time. Because the most estimates are for 2000 (65), 2010 (97), 2050 (152), 2070 (148) and 2100 (121) (Fig. 3c), we selected these five target years as key reference points to capture the trend of both historical and future estimations.
- (2) Interval. Because the evolution of uncertainty is assessed diachronically within predefined time periods (for example, ‘has the uncertainty in the estimation for 2020 decreased based on studies published between 1990 and 2000, 2000 and 2010 and 2010 and 2020?’), the interval length of the published studies determines how trends are detected. To examine the impact of the selection of the interval period, we explored three options: a 10-year interval (which may capture short-term variability but be fragile against data noise), a 20-year interval (better in capturing long-term trends but potentially overlooking short-term variability) and a 15-year interval to balance both options.
- (3) Target-year interval. In some cases the estimation of irrigation water withdrawals was not done for a specific year but for a temporal range (for example, for 1981–2015⁷³, for 1980–2010⁷⁴), depending on research goals or data availability. To account for this, we allowed some analytical paths to vary by ± 10 years from the specified target year. For instance, if the target year was 2010 and the target-year interval factor was activated, all estimations done for 2000–2020 were included in the analysis. When the target year was 2100, the target-year interval only allowed for 2090–2100.
- (4) Rolling window factor. Fixed intervals assume that changes in time-series data occur at strict time breaks (for example, 1990–2000 in a ten-year interval; 2000–2015 in a 15-year interval). Rolling windows allow to detect patterns in a more gradual way if there is no clear breakpoint at which to time bin the data and hence help explore how uncertainty trends depend on period definition. We allowed for either a 0% or a 50% overlap between time bins: for example, a 10-year interval with a 50% rolling window shifts the periods forward by five years (for example, 1970–1980, 1975–1985, 1980–1990...).
- (5) Metric. Uncertainty is multi-dimensional, with different metrics capturing distinct aspects of spread. To ensure that the inference is not determined by the selected metric, we considered four dimensions of spread: dispersion around the mean (which we appraised with the standard deviation and the variance), relative variability (for which we used the coefficient of variation), distribution shape (which we approached with entropy) and the range (which we approached with the range and with the interquartile range).

We combined all levels for these five factors in a full factorial design, resulting in a matrix with a column dimension of five and a row dimension of $5 \times 2 \times 3 \times 2 \times 6 = 360$ rows reflecting all possible analytical paths. We treated each factor as a trigger that determines how to conduct the analysis (for example, how to split the dataset, which is the target year of the estimation, which metric to use to estimate the uncertainty). For each row, the output were n points (with n equalling the number of time bins resulting from the conditions set in

each specific path) ordered in chronological order and reflecting the spread of IWW estimates as per the metric selected. The final output was either ‘increase’, ‘decrease’ or ‘unstable’ depending on whether the uncertainty increased, decreased or was unstable over time (Fig. 3f provides examples). Ten percent of paths (32) lead to analytical pipelines with only one estimation per time bin and hence were removed from the final analysis.

Random forest classification

To assess how methodological choices shaped perceived uncertainty trends, we trained a random forest model with 5,000 trees and selected three predictors per split (Fig. 2). We trained the model on the dataset containing the 360 experiments after excluding analytical paths producing single-point trends (~10% of paths). Predictors included target year, target-year interval, interval, rolling window factor and metric. To address class imbalance and ensure that underrepresented categories received greater influence during training, we applied weights of 1.5, 1.5 and 1 to the ‘increase’, ‘decrease’ and ‘random’ trend categories. The out-of-bag error rate was 15%, with correct classification rates of 86%, 82% and 85% for ‘increase’, ‘decrease’ and ‘random’ trends, respectively. We conducted the random forest analysis with the R package `randomForest`⁷⁵.

Data availability

The datasets generated by our analysis are available via Zenodo at <https://doi.org/10.5281/zenodo.14710666> (ref. 28) and GitHub (https://github.com/arnalduy/uncertainty_irrigation_modelling).

Code availability

The code to reproduce our results is available via Zenodo at <https://doi.org/10.5281/zenodo.14710666> (ref. 28) and GitHub (https://github.com/arnalduy/uncertainty_irrigation_modelling).

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Author contributions

A.P. designed the conceptual approach, with contributions from C.A.-R. and S.N.L. C.A.-R., S.N.L. and N.C. collected the IWW data. O.R., S.F. and L.A.M. close read the papers for calls to reduce uncertainty and research goals. A.P. conducted the statistical analysis, evaluated UA/SA practices and analysed the data, with contributions from C.A.-R., S.N.L., O.R., S.F. and L.A.M. A.P. wrote the manuscript, with contributions from L.A.M. All authors revised and agreed on the final version of the manuscript.

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Competing interests

The authors declare no competing interests.

Additional information

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